💡 Did You Know?

ER Diagram stands for Entity Relationship Diagram. It is used to visually design the structure of a database before creating it.

📖 What is an ER Diagram?

An Entity Relationship (ER) Diagram is a graphical representation of a database. It shows entities, attributes and relationships between different data objects.

⭐ Why do we use ER Diagrams?

- Design databases easily
- Show relationships clearly
- Reduce database errors
- Improve database planning
- Simplify communication

📚 Components of ER Diagram

Component	Description
Entity	Represents an object.
Attribute	Represents properties of an entity.
Relationship	Connects two or more entities.
Cardinality	Shows the number of relationships.

🏢 Entity

An Entity is a real-world object about which data is stored.

```
Student

Teacher

Course

Book
```

🏷 Attribute

Attributes describe the characteristics of an entity.

```
Student

↓

ID

Name

Department
```

```
CGPA
```

🔗 Relationship

A Relationship connects entities together.

```
Student
↓
Enrolls
↓
Course
```

📊 Cardinality

Cardinality specifies how many entities participate in a relationship.

Type	Meaning

1 : 1	One to One
1 : M	One to Many
M : N	Many to Many

💻 Simple ER Diagram

```
Student

↓

Enrolls

↓

Course
```

⭐ Advantages

- Easy database design
- Clear visualization

- Better planning
- Reduces mistakes
- Easy communication

❌ Common Mistakes

- Confusing entity and attribute
- Wrong cardinality
- Missing relationships
- Using duplicate entities

🎤 Interview Questions

Q1: What is an ER Diagram?

An ER Diagram is a graphical representation of a database.

Q2: What is an Entity?

An Entity is a real-world object stored in a database.

Q3: What is an Attribute?

An Attribute describes an entity.

Q4: What is Cardinality?

Cardinality defines the number of relationships between entities.

⚡ Quick Revision

Entity Object

Attribute Property

Relationship Connection